$$= E(X^2) - 2E(X)\,E(\overline{X}) + E(\overline{X})^2$$

$$= E(X^2) - 2\overline{X}\ \overline{X} + (\overline{X})^2$$

$$= E(X^2) - (\overline{X})^2$$

$$= E(X^2) - [E(X)]^2 \qquad \text{where } \overline{X} \text{ is constant and } E(X) = \overline{X}$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

This is called the 2nd central moment about the mean and is known as the variance of the random variable X.

i.e., $\quad \text{Variance} = \text{Var}(X) = \mu_2 = \mu'_2 - (\mu'_1)^2$

$$\text{Standard Deviation (S.D)} = \sigma = \sqrt{variance}$$

Some results based on variance:

(i) Var $(c) = 0$ i.e. Variance of a constant is zero

(ii) If c is constant then Var $(cX) = c^2$ Var(X)

(iii) If X is a random variable and c is a constant then Var $(X \pm c)$ = Var (X)

(iv) a and b are constants then Var $(aX \pm b) = a^2$ Var (X)

(v) a and b are constants then Var $(a \pm bX) = b^2$ Var (X).

(vi) If X and Y are independent random variables then Var $(X + Y)$
= Var (X) + Var(Y)

9.8.3. Moment generating function(M.G.F.)

Definition:

A moment generating function (m.g.f) of a random variable X about the origin is denoted by $M_x(t)$ and is given by

$M_x(t) = E(e^{tx})$, $|t| < 1$

(i) $M_x(t) = \sum(e^{tx})\,p(x)$ for a discrete distribution

(ii) $M_x(t) = \int_{-\infty}^{\infty} e^{tx} f(x)\,dx$ for a continuous distribution

For a random variable X to find the moment about origin we use moment generating function.